

8 The equation of a curve is $y = 5e^{-\frac{1}{2}x} + e^{\frac{1}{2}x}$.

- (a)** Find the coordinates of the stationary point, in logarithmic and/or surd form where appropriate. [5]

- (b) Determine the nature of the stationary point of the curve. [2]

- (c) Explain why the gradient is an increasing function for all real values of x . [2]